

Impact of Traffic Load and Network Parameters on IEEE 802.11 Wireless LAN Performance



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Objectives & Evaluation Framework

Core Objective: Evaluate IEEE 802.11 WLAN performance under varying host density, contention, channel interference, and medium architecture, using discrete-event simulation in OMNeT++/INET.



Task 1 — MAC-Layer Saturation

Network saturation driven by CSMA/CA exponential-backoff retransmissions as host density rises.



Task 2 — Channel Interference

Physical-layer link degradation under variable background noise (SNIR vs. BER).



Task 3 — Wired vs. Wireless

Comparative benchmarking of wired Ethernet and wireless 802.11 under identical saturation load.

EVALUATED METRICS

PDR

Throughput

CWmin

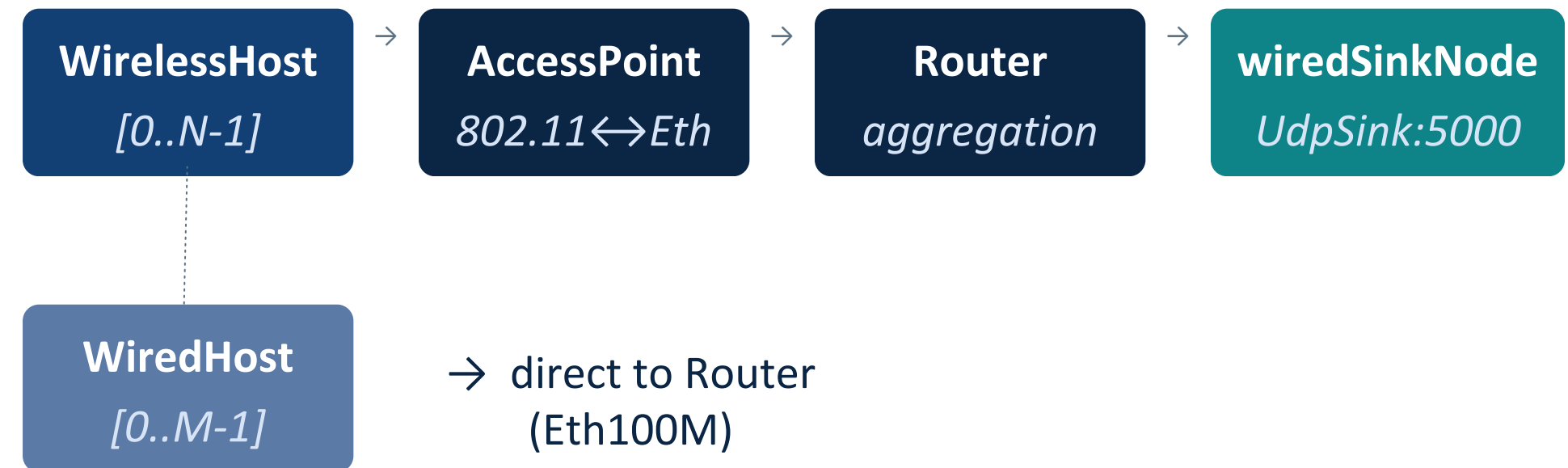
End-to-End Delay

BER

Network Architecture & Simulation Topology

- **Wireless Segment:** N stationary hosts in a $400\text{ m} \times 400\text{ m}$ grid; RadioMedium background noise floor of -10 dBm ; Ieee80211YansErrorModel.
- **Access Point (ap):** Bridges 802.11 wireless frames directly onto the wired Ethernet segment.
- **Wired Segment:** M WiredHost nodes connected via full-duplex 100 Mbps Ethernet (Eth100M).
- **Traffic Destination:** Unidirectional UDP flow to wiredSinkNode, listening on UDP port 5000.

SIMULATION TOPOLOGY — WiredAndWirelessHostsWithAP.ned



All traffic converges on a single UDP sink, so wired and wireless segments are benchmarked under identical routing and traffic-generation conditions.

Task 1 — MAC Contention & Network Saturation

Offered Load: $Cumulative\ Load\ (bps) = N \times Packet\ Size\ (bits) / Send\ Interval\ (s)$ — 8 Mbps per host at a 0.001 s interval.

Low Load (N = 2, 16 Mbps)

Minimal contention.
PDR reaches 92.30%
Delay just 15.71 ms.

Medium Load (N = 4–8)

Demand nears physical limits.
Throughput peaks at N = 8
(31.50 Mbps) as PDR falls to 60.45%.

Saturation Knee (N ≥ 8)

Exponential backoff doubling.
CW_{min}: 15→31→63→960
PDR collapses to 18.00%.

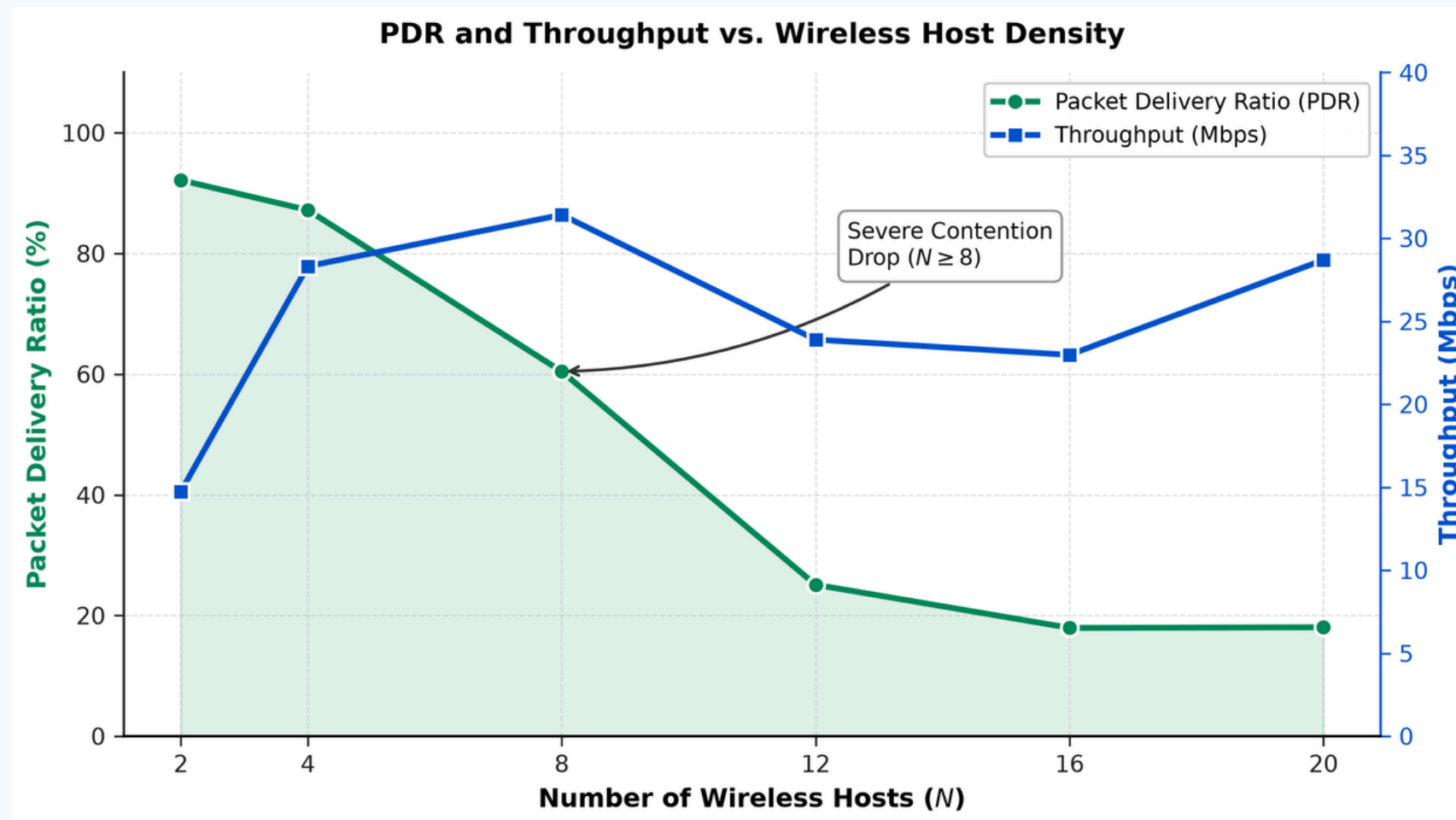


Figure 1. PDR & Throughput vs. Wireless Host Density

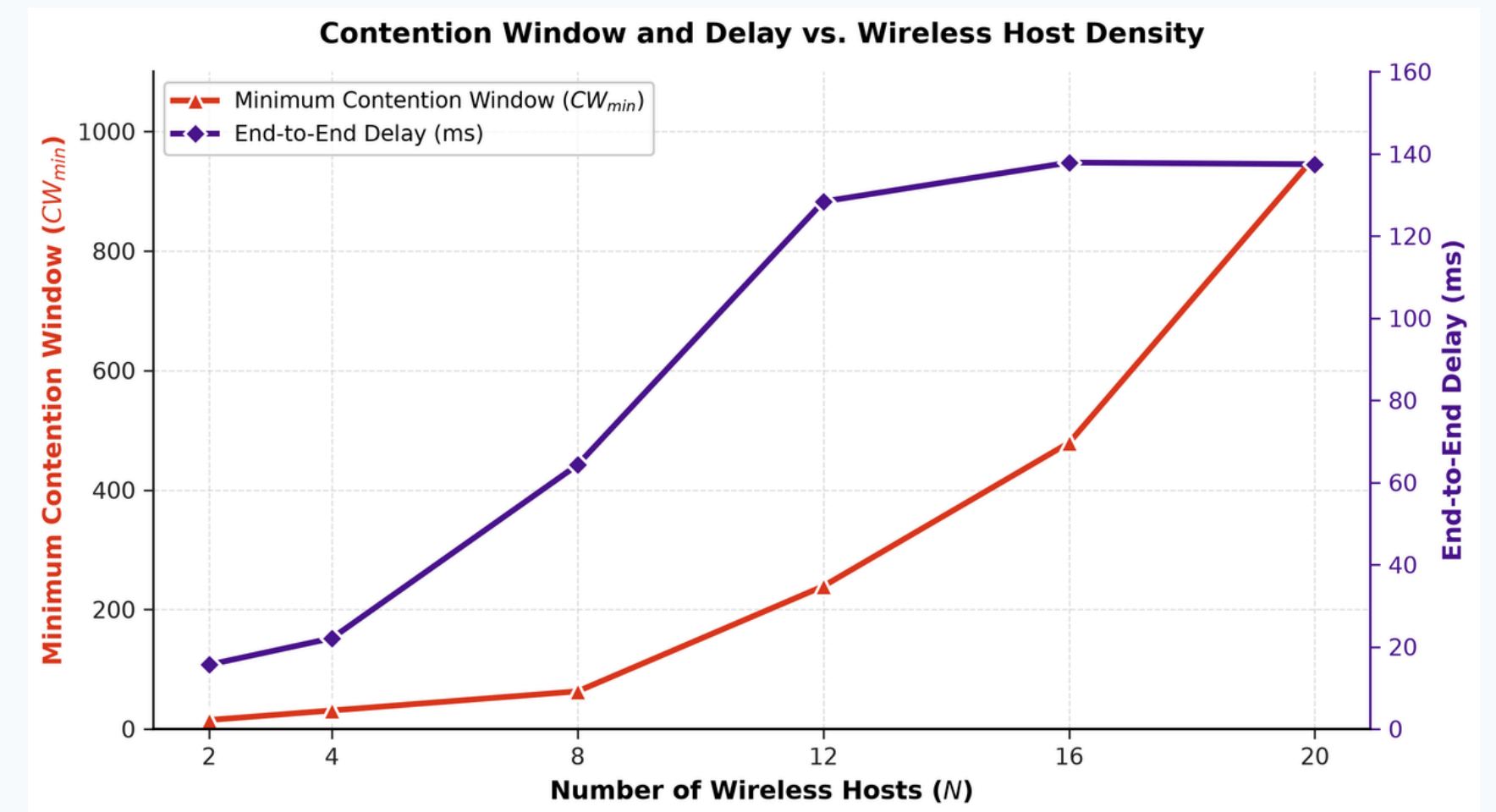


Figure 2. Minimum Contention Window & Delay vs. Host Density

Task 2 — Channel Interference & SNIR Analysis

Setup: Fixed N = 10 hosts (~40 Mbps offered load); background noise swept so SNIR ranges from 30 dB down to 2 dB.

$$BER_{total} = [(BER_{hdr} \times L_{hdr}) + (BER_{data} \times L_{data})] / (L_{hdr} + L_{data})$$



Optimal Region (SNIR ≥ 15 dB)

Near-zero BER; 100% PDR; throughput maxed at 31.50 Mbps.



Critical Degradation (SNIR < 15 dB)

Small per-bit error rises are amplified exponentially; frame success crosses the 50% PDR threshold.



Complete Failure (SNIR = 2 dB)

FCS failures dominate; PDR = 0.00%; delay hits the 500 ms retry ceiling.

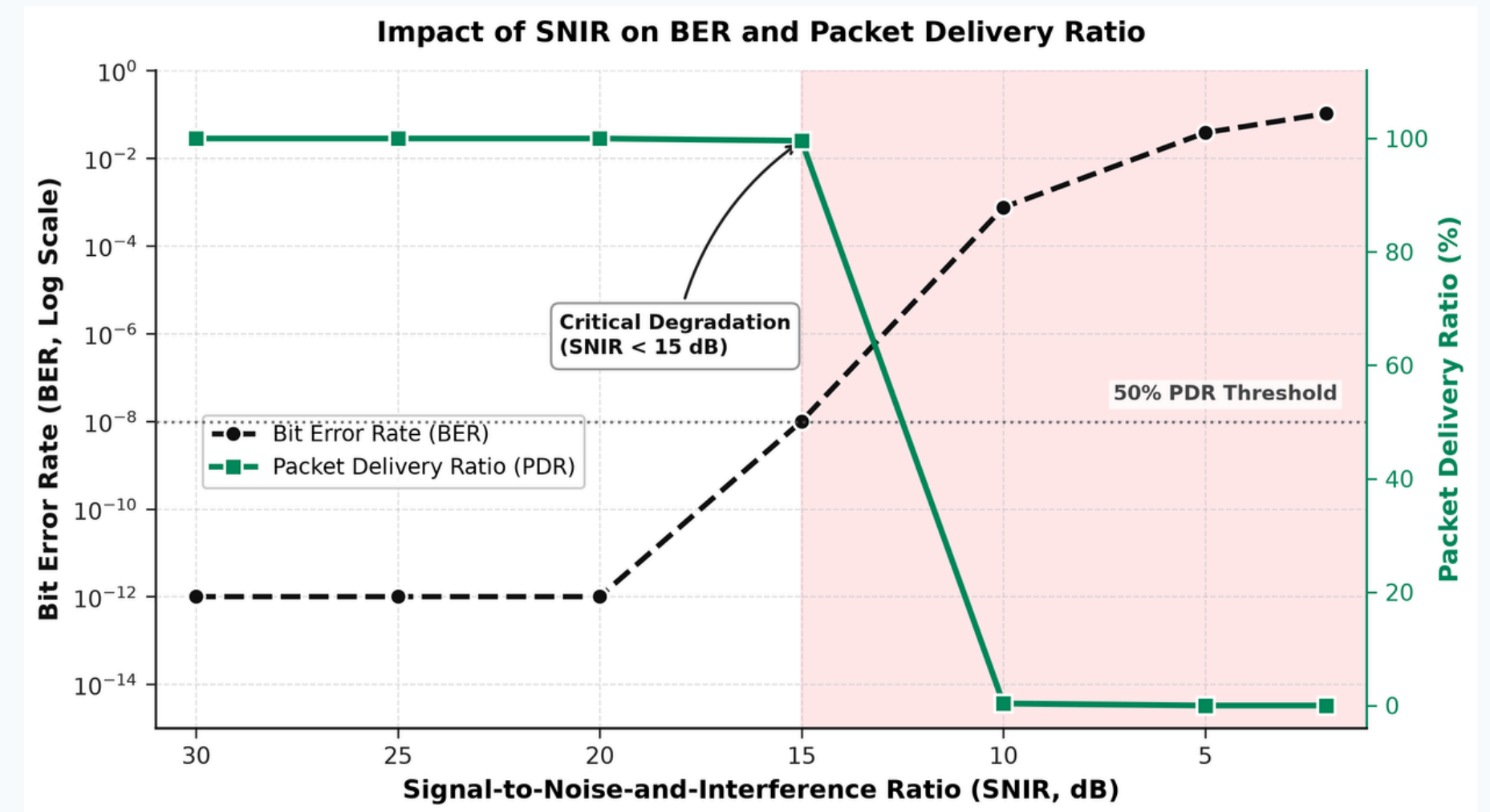


Figure 3. Impact of SNIR on Bit Error Rate and Packet Delivery Ratio

Task 3 — Wired vs. Wireless Saturation Comparison

Setup: Equivalent 160 Mbps offered load targeting a 100 Mbps bottleneck link, at 90%, 100%, and 110% overload factors.

Performance Characteristic	Wireless (802.11)	Wired (Ethernet)
Media Access Control	CSMA/CA (Half-Duplex)	Switched Ethernet (Full-Duplex)
Collision Overhead	High — wasted airtime on retries	Zero — isolated collision domains
100% Load PDR	42.10%	99.20%
100% Load Throughput	29.80 Mbps	95.10 Mbps
100% Load Delay	88.40 ms	1.20 ms

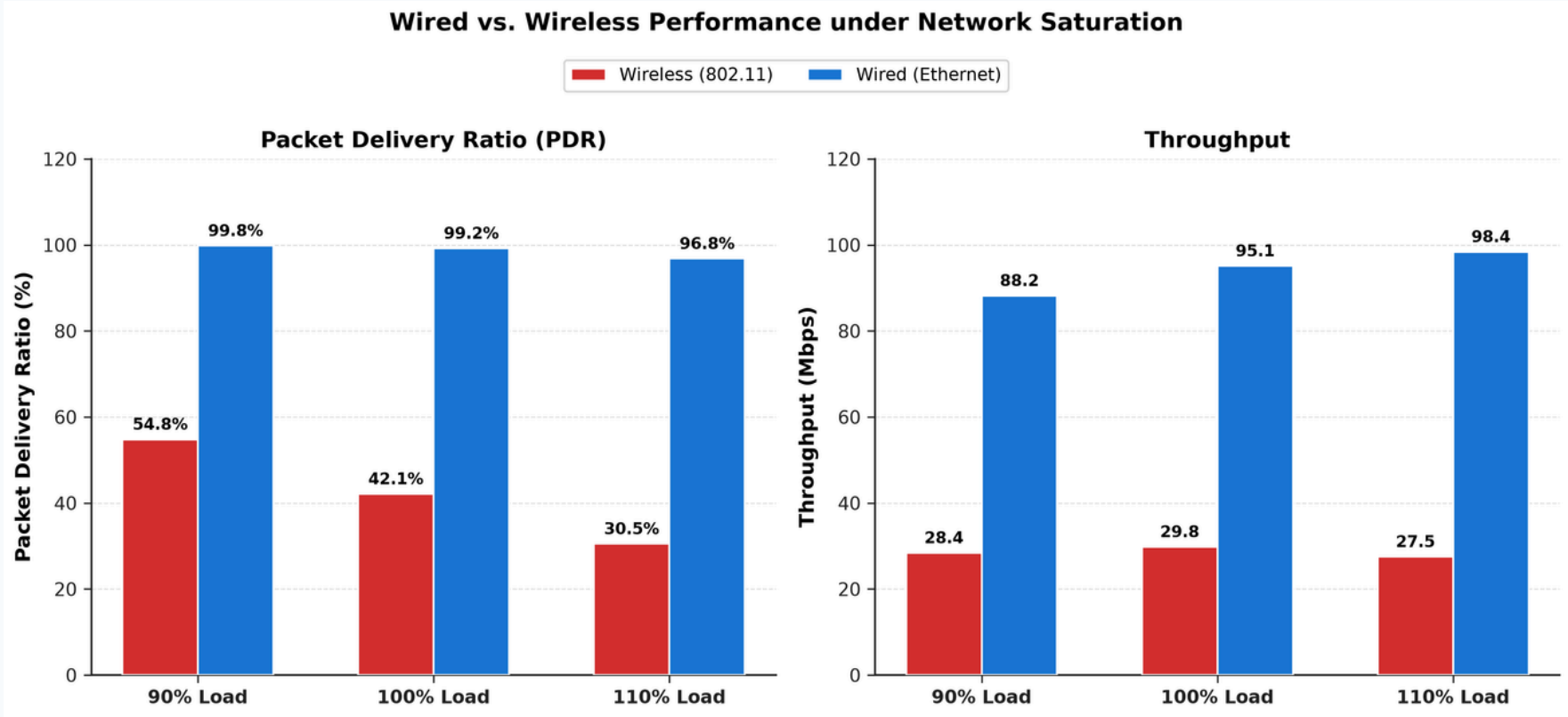


Figure 4. PDR and Throughput: Wired vs. Wireless under Saturation

Wireless capacity is bounded well below the wired ceiling at every load factor tested — the gap widens further under 110% overload.

Overall Statistical Summary

Aggregated across all simulation runs in Tasks 1–3.

PDR

62.45%

average

RANGE

0.00% – 100.00%

PRIMARY DRIVER

*MAC collisions &
background noise
corruption*

THROUGHPUT

32.10

Mbps
average

RANGE

0.00 – 95.40 Mbps

PRIMARY DRIVER

*Physical link capacity &
CSMA overhead*

END-TO-END DELAY

185.30

ms
average

RANGE

0.85 – 1420.50 ms

PRIMARY DRIVER

*Retransmissions & queue
backoff accumulation*

BIT ERROR RATE

0.0834

average

RANGE

0.0000 – 0.4821

PRIMARY DRIVER

*Physical noise floor
degradation*

Key Conclusions & Recommendations



Collision Mitigation in High Density ($N \geq 8$)

Enable RTS/CTS handshaking or increase the initial contention window ($CW_{min} = 63$) to prevent collision cascades.



Dynamic Link Adaptation

Implement adaptive Modulation and Coding Schemes (MCS) to maintain link stability when SNIR drops below 15 dB.



Network Planning

Use multi-channel Access Point deployments and 802.11e EDCA traffic prioritization to eliminate half-duplex wireless bottlenecks.

THANK YOU